Status consumption under Income Inequality

May 12, 2025

Abstract

Status motivations for consumption have generated an immense interest at a time when social mobility has been relatively stagnant - but the consequences of inequality on consumption are still poorly understood. To explain status motivations under environments of income inequality, we treat opportunities for mobility as a link between economic and social goals of the individual and view status consumption as investments towards future economic success. Considering an economy with a dynamic hierarchy of rich and poor positions that participants are able to move between, we then ask how relative incomes may influence the agents' consumption. With a diminishing returns to wealth utility optimised by participants endowed with randomly distributed success-determining attributes, we find that a rise in income inequality provides more incentive to participate in status competitions. As participants bear costs for the mobility, the long-term equilibrium conditions suggest two states - one where both rich and poor strive for status and the other where only the rich participate in status competitions.

1 Introduction

The consumer response to relative income has appealed to economists since at least as far back as Veblen (1899). While the post-war improvements in life-standards (Hirsch, 1977; Frank, 1993) have renewed an interest in positional concerns, the effects of economic inequality on status consumption remain poorly understood. On one hand, the post-war literature on status competitions attributes higher status consumption to a fall in income inequalities and improvement in life-standards (Hirsch, 1977; Galbraith, 1994, 1958; Roychowdhury, 2016) but on the other hand, empirical studies continue to associate more status consumption with higher economic inequality (Nieftagodien, Van Der Berg, et al., 2007; Mazzocco, Rucker, Galinsky, & Anderson, 2012; Chai, Kaus, & Kiedaisch, 2019; J. J. B. Mijs, 2019; J. J. Mijs & Savage, 2020).

The literature in sociology (see Wegener (1992) for a discussion) suggests that one reason why such contradictions arise is that our attributions of inequality to status-differences do not always reconcile the subjective aspects of status (the effect of status differences on individual behaviour) with its objective aspects (the observed reality of status differences). The paper attempts to address this issue by assuming a specific link between economic and social goals and thus by allowing status to have an indirect (instrumental) effect on consumer wealth (Cole, Mailath, & Postlewaite, 1992; Postlewaite, 1998). Restricting our discussion to social capital investments that increase the chances for economic mobility (e.g. education, club memberships), we view status competitions as games that provide direct gains to the participants as probabilistic economic rewards pertaining to status.

The relationship between economic and social goals of an individual are delineated using probability functions that determine a promotion to a richer economic class. The probability of promotion for the participant is driven both by her social capital investment and a set of fixed individual attributes. The status of the participant comes only from her accumulated wealth and is not directly factored into the utility (an approach similar to Cole et al. (1992); Postlewaite (1998)). In the probabilistic game thus defined, the participants compete both with raw talent and acquired skills (social capital) to get promoted to (or to maintain) a rich position. More specifically, the winner of such a game is promoted to a rich position (or maintains her rich position) by having the maximum total score comprising both of raw-talent r_i (uniformly distributed in a population) and social capital s_i (enhanced with expenditure). The social-capital s_i that is enhanced through expenditure is also subject to budget constraints imposed on the participant.

The game comprises only of two types of participants in the above view of status - the poor and the rich - so that the investments towards status (such as club-membership fees, or higher-education) increase the chances to be promoted to the rich status. The rich participants in the game or those who benefit with a higher disposable income (y_2) solely on grounds of their rich status while the poor participants are those who receive a disposable income $y_1 < y_2$. Those who don't participate in the tournament certainly suffer with being poor, but those who do spend only benefit probabilistically.

A useful analogy to explain such a competition is that of tennis leagues or golf tournaments - where rewards are risky but are unlike lotteries (i.e. more expenditure does not linearly increase one's chance to win). As with the tennis or golf, participants with appropriate raw-talent and skills may exist naturally in a population but their success also depends on institutional engagement with other players. More specifically, there may exist biological characteristics such as strength, dexterity, height or psychological abilities (resilience, conscientiousness etc.) that help one's success, but there also exist several characteristics such as technique, rankings, knowledge of equipment etc. that are only available at a significant cost to a participant. The enhancement of these latter skills (social capital) are what the expenditure-depending institutional support are meant to represent in the above probabilistic game. Several real-life status competitions are similar to such tournaments where interaction is small-scale as well as personal. Thus, social capital investments have a strong role to play in such real-life games - interviews, formal introductions or similar business interactions.

Instead of population-wide tournaments, these are small-scale competitions that vary the criteria of success across multiple stages. The competitions are thus meant to be independent and vary only in the threshold participation cost and range of distribution of r_i, s_i of the participants. The presented model considers a single instance of a competition with known distribution of r_i and s_i among a relatively small number of participants who exercise their choice to spend on social capital without an *a priori* knowledge of r_i and s_i.

A few properties of the game are worth highlighting. First, only one player with the maximum score can win a particular tournament. Second, the rank of the participant (rich / poor status) is of no direct significance utility from wealth. Lastly, the raw-talent r_i and skills s_i are of no significance to expenditure decisions - which vary solely on the basis of rich or poor status. A closed-form solution with a diminishing utility from wealth and a threshold level expenditure required for participants to spend in order to avail social capital suggests that the only two equilibria relevant for status consumption are one where rich as well as poor participant participates in the status competitions and one where only the rich participant engage in competitions. A solution where only the poor participate in status consumption can be easily ruled out in the above framework. Further, we also find that higher income differences encourage both poor and rich participants to engage in status competitions while high participation costs discourage the participation of the poor participant.

The result from the model help us explain how higher income differences and participation costs material to status could contribute to a rise or decline in social investments in education or network-building activities in a society. More particularly, the model shows how social capital investments are subject to a discouragement through high participation costs but are still incentivised in an environment of higher income differences.

2 Literature Survey

While status is often contrasted as a social goal rather than the economic goal of an individual, the difference between social and economic goals remains unclear in both sociology and economics (Wegener, 1992). The idea that status is orthogonal to economic goals - or the "pure status" view - seldom fits with the real-world where status both contributes to income and improves with rises in income (Moldovanu, Sela, & Shi, 2007). Indeed the other extreme view that the economic goals alone motivate status competitions - so that status is derived purely from economic rewards is also unrealistic. As Moldovanu et al. (2007) argue, the reality is often somewhere in between the two extremes. Truyts (2010) go further in saying that status would in fact be of little importance if there were no interdependence between status and consumption or labour decisions.

A structural model for the interplay between economic and social goals is however yet to specified in either sociology or economics (Truyts, 2010). The dominant view for status in both sociology and economics has been based on hierarchy and social deference (Coleman, 1990; Goldthorpe, 2000) - a view that is inherent in occupational status surveys widely used in empirical studies (e.g., the EGP classes or the Goldthorpe class scheme). A fundamental problem that the sociology literature has elaborated (Wegener, 1992) is the failure to reconcile the difference between objective aspects of status (the observed reality of status differences) and its subjective aspects (the effect of status differences on individual behaviour). An evident issue with addressing only the objective aspect of status using a utilitarian model while leaving the subjective aspect exogenous is that status may never be explained in economic terms when an economy for status is ruled out. While discussions on status often rely on a deference-based view of status, the subjective (private) evaluations of status in empirical studies measure an overall desirability of occupations or education instead (Goldthorpe & Hope, 1974; Adler & Kraus, 1985) .

The issue also makes it difficult to answer whether inequalities can have a negative or positive effect on status consumption or not. On one hand, the post-war literature on status competitions argues that a fall in income inequalities and improvement in life-standards has resulted in a higher focus on status goods (Hirsch, 1977; Galbraith, 1958, 1994, 2001). On the other hand, the empirical literature often finds status consumption to increase with rising inequality. In a study focusing on education expenses for status purposes in China, Jin, Li, and Wu (2011) find, for example, that higher inequality increases the consumption towards status. Similarly, Jaikumar and Sarin (2015) also report that consumers in India have increased conspicuous consumption with rise in inequality. An appropriate line of research has been to examine the environment where inequality is experienced (often relying on social attitudes surveys). Roychowdhury (2016) measure a visible inequality based on reference groups to argue that reduction of visible inequalities does increase status consumption. J. J. B. Mijs (2019) use the data from International Social Survey (Haller, Jowell, & Smith, 2009) to detail how a rise in inequality might have cemented a belief in meritocracy in the developed economies and coincided with segregation rather than a healthy competition. Since the environment where inequality is experienced comprises both of the objective and subjective notions of status, a link between the two provides us a model to better understand the effects of inequality on status consumption.

The presented approach specifies a particular link between social and economic goals by considering how certain characteristics are nurtured or discouraged by social institutions (Goode, 1978). This is contrasted against notions of status based on tradeable-power in an economy (Coleman, 1990) - a view that has also been criticised due to its inherent assumption of there being a fixed total amount of status in society (Wrong, 1979). Instead of using status as a wealth-determined goal, we follow an approach suggested by Postlewaite (1998) to consider status only of an indirect role in maximisation of wealth utility. More specifically, the rank of the participant (rich or poor status) does not directly contribute to utility and but matters only through a higher income that contributes to maximisation of the participant's expected wealth utility.

In summary, the view of status as an economic goal that corresponds to a possible future monetary benefit to the participant - rather than a non-monetary utility - is unlike the view of status signaling used in many studies on conspicuous consumption (Heffetz, 2011; Ireland, 1994; Omori & Smith, 2015; Charles, Hurst, & Roussanov, 2009; Kaus, 2013; Khamis, Prakash, & Siddique, 2012; Friehe & Mechtel, 2014; Jaikumar & Sarin, 2015; Knell, 1999; Falk & Knell, 2004). This is primarily because the utility from status consumption in the empirical studies on conspicuous consumption is treated both necessarily futile and entirely separable from a private (non-visible) utility. Our specific consideration of status - an approach that equates the social goals of the participant with her long-term economic goals - bears more similarity with the approach followed by Cole et al. (1992) where status does not directly contribute to participant's utility.

3 Model

In the status game that selects the winner to be maintained at the rich position, only two characteristics matter for every individual. The first is the social capital that can be enhanced with expenditure by the individual participant and the second is a raw-talent that cannot be changed with higher expenditure. Both skill s_i (enhanced with expenditure) and the raw talent r_i (which cannot be changed with higher expenditure) are assumed to be uniformly distributed in the population. Assuming the weight μ of raw-talent in the maximum total score considered for the win in a competition, we consider the preferences for the probability of $\mu r_i + (1-\mu)s_i$ (where r_i is raw-talent and s_i are skills that are available with higher expenditure) under the assumption that the population remains constant over time (there are the same number of rich and poor in total) and that the participants are not aware of their r_i, s_i before participating in the competitions. In other words, the rich participants are replaced by either a rich or poor participant only through the tournament that selects highest combined score regardless of rich or poor background of the participant. Since all participants can participate in every such tournament, it is possible for a rich participant to lose her rich-position and let another participant acquire her position.

We also assume that the preferences for all poor participants are the same as are the preferences of all the rich participants. Having a rich or a poor status by itself only has an indirect effect (through higher expenditure or budget) on the chance to be promoted (or maintaining one's rich position) - a probability that otherwise depends on the individual ability r_i and social-capital s_i. All rich participants and all poor participants having the same behaviour also means that the individual attributes r_i and s_i - which are not known to the participant - are not used to decide upon the expenditure (effort) towards the status game.

With r_i and s_i uniformly distributed, we can formulate the likelihood of any participant $i \in [1, N]$ as the probability for their total score $\mu r_i + (1-\mu)s_i$ being higher than any other participant - where $\mu \in (0,1)$ is a parameter that combines social-capital s and raw-talent r so that the participant with maximum score $\mu r + (1-\mu)s$ wins the competition. The winner of the competition acquires a rich position so that she obtains the disposable income y_2 while the loser acquires the poor position receive $y_1 < y_2$.

To start with a simpler case, consider that in a competition with raw-talents alone, the winning participant is one who has the maximum r_i drawn from the sample N. In other words, the winner in a competition of raw-talents is a player j such that $\{r_i < r_j \forall i \in [1, N]\}$ and the probability that a given participant i of raw-skill r_i is the winner is essentially that of all other players $j \neq i$ having ability $r_{j \neq i}$ less than r_i. This is the probability

$$\int_0^1 P(r_i > r_1) \times P(r_i > r_2) \times ...P(r_i > r_{i-1}) \times P(r_i > r_{i+1})...P(r > r_N)dr_i$$

With a uniform distribution of skills $r \in [0,1]$, we have $P(r_i \geq r) = r_i$ for all $i \in [1, N]$ and therefore, the probability is evaluates to $\int_0^1 (r_i)^{N-1} dr_i = \frac{1}{N}$. With more players, the chances of a particular participant winning evidently declines.

Introducing the skills s_i which are curated in social institutions, we consider a threshold investment $\overline{\nu}$ which is necessary to be spent for the social capital to be available to a participant. While this is an arrangement that evidently favours the rich participant, there are also diminishing returns to investing $\overline{\nu}$ towards the richer position if naturally occurring raw talent r_i is also significant in the competition. To direct our focus on conditions when social capital is more important than raw-talent, we assume in the foregoing analysis that raw-talent is less valued than skills cultivated by institutions[1]. As a result, the condition $1 - \mu > \mu$ (which is equivalent to the following assumption) has been used throughout in the rest of the discussion.

Assumption 1. $\mu < \frac{1}{2}$

With the threshold investment $\overline{\nu}$ as a discrete response to costs of participation, the participants face different probabilities to win depending on whether they spend $\overline{\nu}$ or not. More specifically, the participant takes advantage of the skills with weight $1 - \mu$ only when she spends an amount $\overline{\nu}$. In the all-or-nothing payment scheme, the participant might avoid the investment altogether and keep the investment towards savings instead. The behaviour of all participants depends only on ν_1, ν_2 - as they are unaware of r_i, s_i before they participate.

The result we use depends on the following distribution function of $x_i \equiv \mu r_i + (1 - \mu)s_i$ under the Assumption 1 where r_i and s_i are independent random variables uniformly distributed between 0 and 1 (see Appendix A) .

[1]While effects of changes in μ over time on the dynamics of the game are of interest in their own right, we have narrowed our interest on a situations were social capital is weighted higher than raw talent.

$$F_X(x) = P(x_k < x) = P(r_k\mu + s_k(1-\mu) < x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{x-\mu/2}{1-\mu} & \mu < x < 1-\mu \\ 1 - \frac{(x-1)^2}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

In the game where a participant i with the maximum score x_i is the winner, the number of rich positions is fixed to $N-P$ so that only the top-scorer gets to occupy the only rich position available to be filled in every competition. The expected behaviour of the rich and the poor participants in the long-term can be explained with four different cases.

- Case I - The poor participant does not participate due to insufficient investment $\nu_1 = 0$ while the rich participant does participate by spending $\nu_2 = \overline{\nu}$.
- Case II - The rich participant does not participate with $\nu_2 = 0$ but the poor participant does participate i.e. $\nu_1 = \overline{\nu}$.
- Case III - Both poor and rich participants do not participate so that $\nu_1 = \nu_2 = 0$.
- Case IV - Both poor rich and participant participate with $\nu_1 = \nu_2 = \overline{\nu}$.

With Case I, the poor participant can win when her raw-talent (weighted with μ) is higher than the total score of all other poor participants as well as the total score of the rich participants (who compete with acquired skills). The poor are priced out of accessing any skills s in their score in Case I so that only the rich participants with investment $\nu_2 = \overline{\nu}$ have skills $s > 0$. Thus, a given poor participant must be ahead both of all of the poor participants and the rich participants so that a participant i may win only if μr_i is higher than than raw-talent μr_k in all poor participants and also exceeds the score $\mu r_k + (1-\mu)s_k$ involving skills among the rich participants

$$P_1(\text{win}|r_i) = \prod_{k=1,k\neq i}^{P} P(\mu r_k < \mu r_i) \prod_{k=P+1}^{N} P(\mu r_k + (1-\mu)s_k < \mu r_i) = r_i^{P-1} F_X(\mu r_i)^{N-P} \tag{1}$$

Therefore,

$$P_1(\text{win}|r_i) = \begin{cases} \frac{\mu^{N-P} r_i^{2N-P-1}}{2\mu(1-\mu)} & 0 < r_i < \mu \\ r_i^{P-1}(\frac{\mu r_i - \mu/2}{1-\mu})^{N-P} & \mu < r_i < 1-\mu \\ r_i^{P-1}(1 - \frac{(\mu r_i - 1)^2}{2\mu(1-\mu)})^{N-P} & 1-\mu < r_i < 1 \\ 0 & \text{otherwise} \end{cases}$$

With the uniform density for r_i (i.e. $f(r_i) = 1$), total probability to win can be calculated as$\int_{r_i=0}^{r_i=1} P(\text{win}|r_i) dr_i$. Since the participants do not know the values of r_i,s_i before they participate in such competitions, the total probability can be evaluated with the assumption that both r_i and s_i range from 0 to 1. Notice also that since only one participant can win the skills-game, the probability $\int_0^1 \int_0^1 P_2(\text{win}|r_i, s_i) ds_i dr_i$ for a rich participant to win is simply $\frac{1}{N-P} - \frac{P}{N-P} \int_0^1 \int_0^1 P_1(\text{win}|r_i, s_i) ds_i dr_i$.

In Case II, when the rich person has no access to social capital (i.e. all rich participants have $\nu_2 = 0$), the poor participant has the skills-advantage in the game so the probability to win for the poor participant i is

$$P(\text{win}|r_i, s_i) = \prod_{k=1,k\neq i}^{P} P(\mu r_k+(1-\mu)s_k < \mu r_i+(1-\mu)s_i) \prod_{k=P+1}^{N} P(\mu r_k < \mu r_i+(1-\mu)s_i)$$
$$= F_X(x_i)^{P-1} \prod_{k=P+1}^{N} P(\mu r_k < \mu r_i + (1-\mu)s_i) \quad (2)$$

In Case III, when neither the rich nor the poor participate in the game, then the game depends only on raw-talents and all participants have the same probability $\frac{1}{N}$ to win the game. This is because $\int_{r_i=0}^{r_i=1} P(\text{win}|r_i)dr_i = \frac{r_i^N}{N}|_0^1 = \frac{1}{N}$ where $P(\text{win}|r_i) = \prod_{k=1,k\neq i}^{P} P(r_k < r_i) \prod_{k=P+1}^{N} P(r_k < r_i) = {r_i}^{N-1}$.

In Case IV, when both rich and poor participants set $\nu_1 = \nu_2 = \overline{\nu}$, then the poor participant can win only if her score x_i is above all other participants - rich or poor.

$$P(\text{win}|r_i) = \prod_{k=1,k\neq i}^{P} P(\mu r_k+(1-\mu)s_k < \mu r_i+(1-\mu)s_i) \prod_{k=P+1}^{N} P(\mu r_k+(1-\mu)s_k < \mu r_i+(1-\mu)s_i)$$
$$= F_X(x_i)^{N-1} \quad (3)$$

Using symmetry arguments, this winning probability - which is the same for all participants - also turns out to be $1/N$ for all the participants.

Notice that with the rich and poor participants having disparate incomes y_1 and y_2 (so that $y_1 < y_2$), the above setup evidently puts the poor participants at a disadvantage since they may not afford to pay the costs to enhance their skills (i.e. realise the true worth of their skills s_i due to high participation costs). However, since both raw-talent r_i and social-capital s_i (the latter enhanced by costly institutional support) are randomly distributed in the population, the competition may never rule out the chance for a poor participant to win. For the same reason, the returns from investments to enhance one's skills may also be diminishing since one cannot beat raw-talent and non-enhanced skills after a certain point.

It is worth emphasising also that the effect of expenditure on the winning probability is weakened with higher number of participants in the game where abilities (both skills s_i relevant for social capital and the raw-talent r_i) are uniformly distributed. In other words, if poor and rich participants were to be considered as two separate groups, then the effect of expenditure on their wins declines when more and more participants are sampled with varied raw-talents and skills. Figures 1 and 2 demonstrate this effect for probability calculated with Monte Carlo simulations for increasing values of the number of poor participants (P) and the total number of participants (N) while hold the ratio P/N constant ($P/N = 1/2$). Our Monte Carlo simulations suggest that the role of expenditure is insignificant for $N \geq 30$. We now explore the effect of income differences using using closed-form solution for $N = 2$ and $P = 1$ in Section 3.1. Equivalent results can be derived using Monte-Carlo simulation for higher values of N and P.

3.1 Equilibrium Conditions

We proceed with a solution by setting total number of participants $N = 2$ and the number

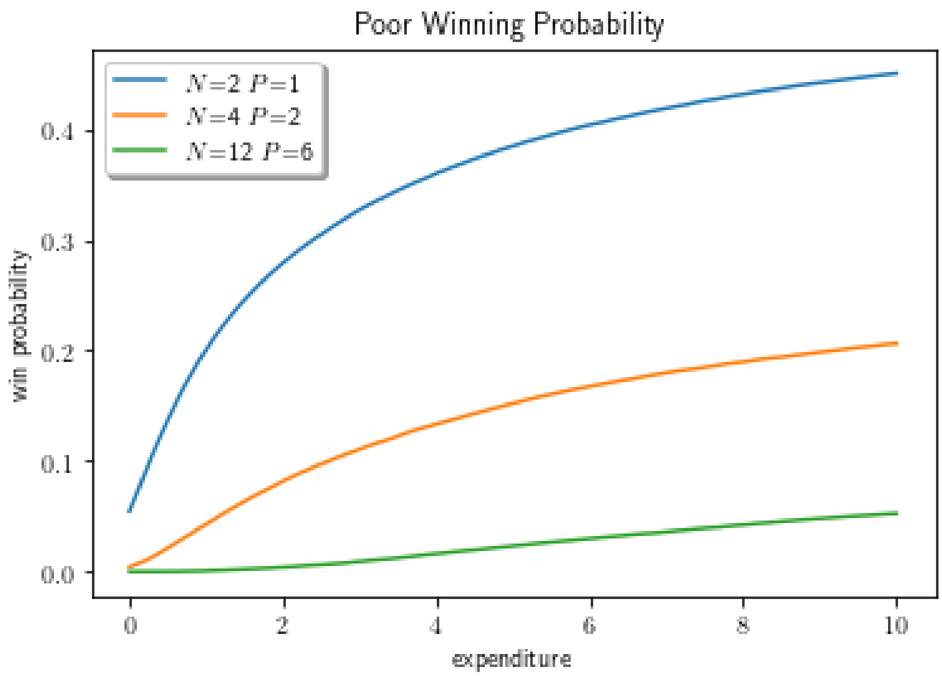

Figure 1: Probability to win for poor participant with increasing number of participants

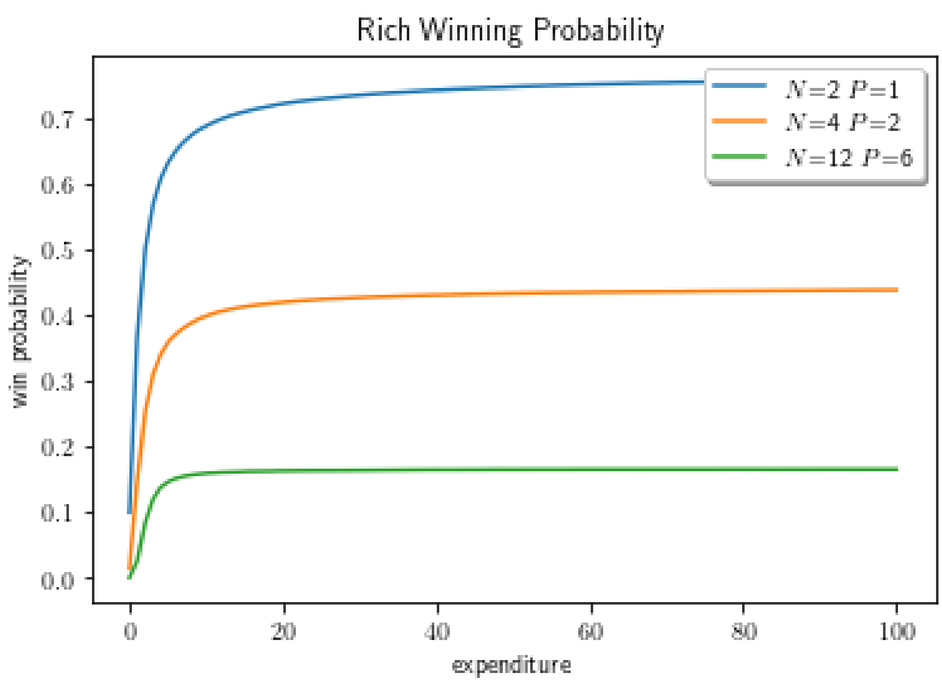

Figure 2: Probability to win for rich participant with increasing number of participants

of poor participants $P = 1$ in Equations 1, 2 and 3 above. As before, a participant takes advantage of the skills with weight $1-\mu$ only by spending an amount $\overline{\nu}$ and can decide to avoid the expenditure altogether in order to invest it towards the savings instead. We first discuss the different probabilities in Case I, II, III and IV before discussing the role of an additive intertemporal diminishing utility from accumulated wealth. Recall that the behaviour of both participants depends only on ν_1, ν_2 - as they are unaware of r_i, s_i before they participate.

An important result for the closed-form solution is the probability distribution (CDF) of $x_i \equiv \mu r_i + (1-\mu)s_i$ under assumption $\mu < 1-\mu$ (for independent variables r_i and s_i that are uniformly distributed between 0 and 1) that is detailed in Appendix A.

$$F_X(x) = P(x_k < x) = P(r_k\mu + s_k(1-\mu) < x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{x-\mu/2}{1-\mu} & \mu < x < 1-\mu \\ 1 - \frac{(x-1)^2}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

We now evaluate above probabilities under Case I, II, III and IV discussed above.

Case I

In Case I, the poor participant can win when her raw-talent r_1 is higher than of the total score of rich participant (who competes with acquired skills). The poor participant is priced out of accessing any skills s in her score so that the skills $s > 0$ is only true for the rich participant having $\nu_2 = \overline{\nu}$. Thus, the poor participant with raw-talent r_1 can win with the following probability

$$P(\text{win}|r_1) = P(\mu r_2 + (1-\mu)s_2 < \mu r_1) = F_X(\mu r_1)$$

Notice that $\mu r_1 < \mu < 1-\mu < 1$ is implied due to $\mu < \frac{1}{2}$ (and $r_1 \in (0,1)$). The probability that the total score $\mu r_2 + (1-\mu)s_2$ does not exceed μr_1 implies that only the first part of the piece-wise function for $F_X(x)$ needs to be considered. Therefore, using $P(\text{win}|r_1) = \frac{\mu^2 r_1^2}{2\mu(1-\mu)}$, we evaluate $P_1(\text{win})$ as

$$P_1(\text{win}) = \int_0^1 P(\text{win}|r_1)dr_1 = \frac{\mu}{6(1-\mu)}$$

Since only one of the participants can win, the probability $P_2(\text{win})$ for the rich participant to win is simply $1 - P_1(\text{win})$. The following thus applies to all other cases (i.e. Cases II, III and IV).

$$P_2(\text{win}) = 1 - P_1(\text{win}) \tag{4}$$

Case II

In Case II, the rich person does not participate ($\nu_2 = 0$) and both the poor participants have the skills-advantage in the game. A poor participant wins by having the total score higher than the raw-talent of the rich participant (who does not acquire skills with investment). Representing $x_i \equiv \mu r_i + (1-\mu)s_i$, we have the following probability for the poor participant to win

$$P(\text{win}|r_1, s_1) = P(\mu r_2 < \mu r_1 + (1-\mu)s_1)$$

Notice that $P(\mu r_2 < \mu r_1 + (1-\mu)s_1)$ (i.e. $P(r_2 < r_1 + (\frac{1}{\mu} - 1)s_1)$) is 1 after $r_1 + (\frac{1}{\mu} - 1)s_1 > 1$ (or $\mu r_1 + (1-\mu)s_1 > \mu$) and $\frac{1}{\mu}$ in the interval $(0, \mu)$ (this is because $r_2 \in (0, 1)$). In other words,

$$P(r_2 < r_1 + (1/\mu - 1)s_1) = \begin{cases} \frac{x_1}{\mu} & 0 < x_1 < \mu \\ 1 & \text{otherwise} \end{cases}$$

The probability to win for the poor participant can be written as (see Appendix A)

$$P(\text{win}|r_1, s_1) = \begin{cases} \frac{x_1}{\mu} & 0 < x < \mu \\ 1 & \mu < x_1 < 1-\mu \\ 1 & 1-\mu < x_1 < 1 \\ 0 & \text{otherwise} \end{cases}$$

Using double-integral properties (see Appendix B), the total probability to win $P_1(\text{win})$ for a poor participant is evaluated as follows

$$\begin{aligned} P_1(\text{win}) = \int_{s_1=0}^{s_1=1} \int_{r_1=0}^{r_1=1} P(\text{win}|r_1, s_1) ds_1 dr_1 &= \int_{r_1=0}^{r_1=1} \int_{s_1=0}^{s_1=\frac{\mu(1-r_1)}{1-\mu}} \frac{(\mu r_1 + (1-\mu)s_1)}{\mu} ds_1 dr_1 \\ &+ \int_{r_1=0}^{r_1=1} \int_{s_1=\frac{\mu(1-r_1)}{1-\mu}}^{s_1=1-\frac{\mu r_1}{1-\mu}} 1 \, ds_1 dr_1 \\ &+ \int_{r_1=0}^{r_1=1} \int_{s_1=1-\frac{\mu r_1}{1-\mu}}^{s_1=1} 1 \, ds_1 dr_1 \\ &= \frac{\mu}{3(1-\mu)} + \frac{1-2\mu}{1-\mu} + \frac{\mu}{2(1-\mu)} \\ &= \frac{2\mu}{6(1-\mu)} + \frac{6(1-2\mu)}{6(1-\mu)} + \frac{3\mu}{6(1-\mu)} = \frac{6-7\mu}{6(1-\mu)} \end{aligned}$$

The probability $P_2(\text{win})$ for the rich participant to win follows from Equation 4.

Case III

In Case III, when neither the rich nor the poor participate in the game, then the game depends only on raw-talents and both participants have the same probability $\frac{1}{2}$ to win the game. This is because $\int_{r_i=0}^{r_i=1} P(\text{win}|r_i) dr_i = \frac{r_i^2}{2}|_0^1 = \frac{1}{2}$. Both the poor and rich participant have the same total probability $\frac{1}{2}$ to win.

Case IV

In Case IV, when both rich and poor participants set $\nu_1 = \nu_2 = \overline{\nu}$, then the poor participant can win only if her score x_i is above that of the rich participant. Therefore,

$$P(\text{win}|r_1, s_1) = P(\mu r_2 + (1-\mu)s_2 < \mu r_1 + (1-\mu)s_1)$$
$$= F_X(x_1)$$

$$P(\text{win}|r_1, s_1) = \begin{cases} \frac{x_1^2}{2\mu(1-\mu)} & 0 < x_1 < \mu \\ \frac{x_1-\mu/2}{1-\mu} & \mu < x_1 < 1-\mu \\ 1 - \frac{(x_1-1)^2}{2\mu(1-\mu)} & 1-\mu < x_1 < 1 \\ 0 & \text{otherwise} \end{cases}$$

While this integral can be evaluated based on symmetry arguments (since x_i is also randomly distributed), the integral $\int_{s_1=0}^{s_1=1}\int_{r_1=0}^{r_1=1} P(\text{win}|r_1, s_1)dr_1 ds_1$ also evaluates to $\frac{1}{2}$ (see Appendix B)

$$\begin{aligned}\int_{s=0}^{s=1}\int_{r=0}^{r=1} F_X(x_1)ds_1dr_1 &= \int_{r_1=0}^{r_1=1}\int_{s_1=0}^{s_1=\frac{\mu(1-r)}{1-\mu}} \frac{(\mu r_1 + (1-\mu)s_1)^2}{2\mu(1-\mu)}\, ds_1dr_1 \\ &+ \int_{r_1=0}^{r_1=1}\int_{s_1=\frac{\mu(1-r)}{1-\mu}}^{s_1=1-\frac{\mu r}{1-\mu}} (\frac{\mu r_1 + (1-\mu)s_1 - \mu/2}{1-\mu})\, ds_1dr_1 \\ &+ \int_{r_1=0}^{r_1=1}\int_{s_1=1-\frac{\mu r}{1-\mu}}^{s_1=1} (1 - \frac{(\mu r_1 + (1-\mu)s_1 - 1)^2}{2\mu(1-\mu)})\, ds_1dr_1 \\ &= \frac{\mu^2}{8(1-\mu)^2} + \frac{1-2\mu}{2(1-\mu)} + \frac{(4-5\mu)\mu}{8(1-\mu)^2} = \frac{4(1-\mu)^2}{8(1-\mu)^2} = \frac{1}{2}\end{aligned}$$

As is clear from Figure 3 and above results (see summaries in Tables 3 and 4), the condition $\mu < 1/2$ - indicating that curated skills are given more importance that raw-talent in the competitions - means that the poor participant is strictly better off (than facing probability $1/2$) if she participates while the rich participant doesn't participate (Case II). Also, the poor participant is better off participating for $\mu < \frac{1}{2}$ when the rich participant does participate as it raises the probability to $1/2$. That the competition becomes uncompetitive both when threshold $\overline{\nu}$ falls too much (letting both participants participate) or rises too much (letting both participants withdraw) also follows from the properties of the winning probability.

Notice that the rich participant could withdraw and face either $1/2$ or lower probability (when the poor participant participates). By participating, therefore, the rich participant would either face probability $1/2$ (when poor participant also participates) or better (if the poor participant withdraws). Considering the probabilities alone, the rich participant would always participate. If the rich participant always participates, then so long as budget constraints permit, the poor participant is also better off participating since not participating brings her probability to win below $1/2$. If the utility was from probability alone, then both participants would participate and face $1/2$ probability.

However, it's not the probability to win that the participants optimise (maximise). A more accurate representation of the real-world is a game where participants face a finite life-time and utility from accumulated assets instead. We therefore explore the case where the participants optimise an intertemporally additive utility over their life-time. The life-time in the model is represented with two time-horizons - where the second-time horizon is one before the end of

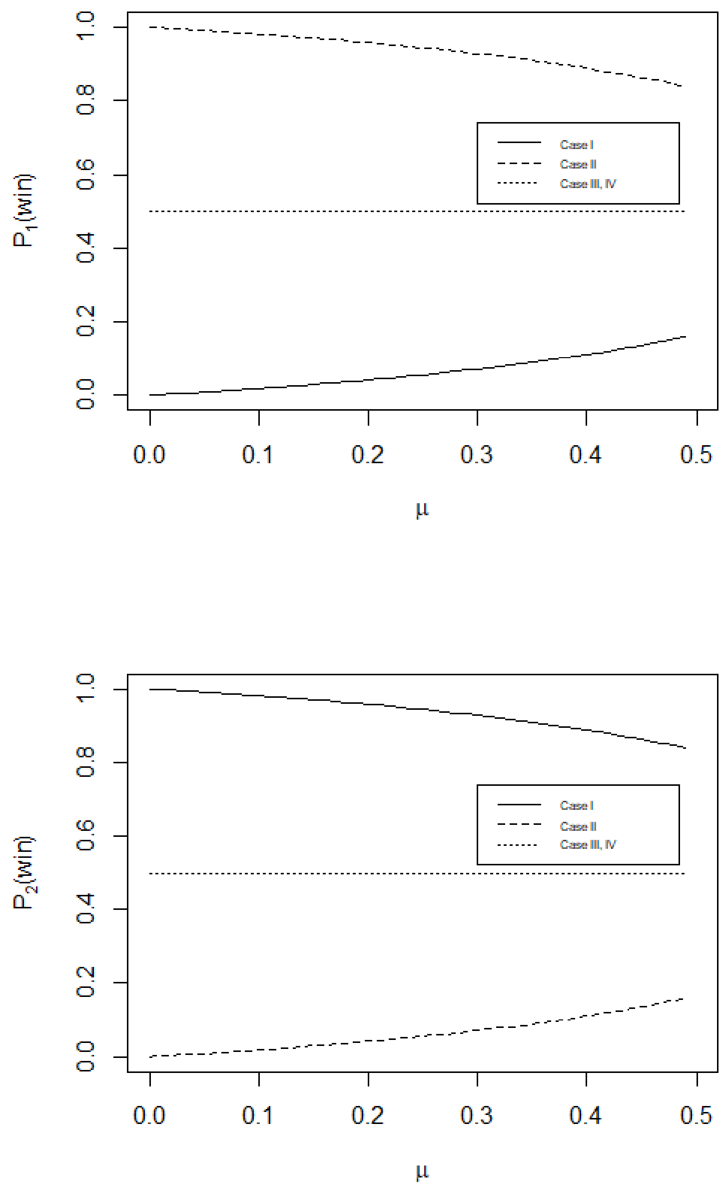

Figure 3: Probability of winning $P_1(\text{win})$, $P_2(\text{win})$ for poor and rich participants with varying μ

life when all wealth accumulated by either participant must disappear. With no accumulations at the end of life, both rich and poor participants must decide to spend all their incomes y_1 or y_2 (for poor and rich participants respectively so that $y_1 < y_2$) towards the promotion game. In the first-stage, however, the participants decide the amount to be spent towards promotion based upon the utility from wealth they obtain across the two time-horizons that span their lifetime. For the poor participant, the total utility to be optimised (assuming EUT) under the intertemporally additive utility $u(\cdot)$, a winning probability function $W(x)$ (for expenditure x) and a discount-factor δ is

$$u(y_1 - \nu_1) + \frac{1}{\delta}((1 - W(y_1))u(y_1) + W(y_1)u(y_2))$$

Similarly, for the rich participant, the total utility to be optimised (assuming EUT) is

$$u(y_2 - \nu_2) + \frac{1}{\delta}((1 - W(y_2))u(y_1) + W(y_2)u(y_2))$$

		Poor *withdraws*	participant *participates*
Rich	*withdraws*	Case III	Case II
Participant	*participates*	Case I	Case IV

Table 1: Cases I,II, III, IV as participant strategies

		Poor participant	
		withdraws	*participates*
Rich participant	*withdraws*	$u(y_2) + \frac{(1-\frac{P}{N})u(y_2) + \frac{P}{N}u(y_1)}{\delta}$ $u(y_1) + \frac{(1-\frac{P}{N})u(y_2) + \frac{P}{N}u(y_1)}{\delta}$	$u(y_2) + \frac{u(y_2)p_2' + (1-p_2')u(y_1)}{\delta}$ $u(y_1 - \overline{\nu}) + \frac{u(y_2)p_1' + (1-p_1')u(y_1)}{\delta}$
	participates	$u(y_2 - \overline{\nu}) + \frac{p_2 u(y_2) + (1-p_2)u(y_1)}{\delta}$ $u(y_1) + \frac{p_1 u(y_2) + (1-p_1)u(y_1)}{\delta}$	$u(y_2 - \overline{\nu}) + \frac{(1-\frac{P}{N})u(y_2) + \frac{P}{N}u(y_1)}{\delta}$ $u(y_1 - \overline{\nu}) + \frac{(1-\frac{P}{N})u(y_2) + \frac{P}{N}u(y_1)}{\delta}$

Table 2: Utility from two periods for (rich,poor) participants - at the start of the first-period (with P poor participants among N total participants)

With above utilities, we discuss conditions for whether the Case I, II, III, IV can become Nash equilibria (see Table 1). We then proceed to discuss the implications for the effect of income differences on expenditure towards status. It is worth remarking that our discussion of the effect of the threshold $\overline{\nu}$ and the income differences $y_2 - y_1$ on participation (expenditure) assumes that the exogenous parameter μ is stable with respect to short-term changes in threshold investment $\overline{\nu}$ as well as disposable incomes y_1, y_2.

For ease of notation we define p_1, p_2, p_1' and p_2' - which correspond to the **two-player** model discussed above.

$$
\begin{aligned}
p_1 &\equiv \frac{\mu}{6(1-\mu)} \\
p_2 &\equiv 1 - p_1 \\
p_1' &\equiv \frac{6-7\mu}{6(1-\mu)} \\
p_2' &\equiv 1 - p_1'
\end{aligned} \tag{5}
$$

The choices for the poor and rich participants using a general intertemporally additive function $u(A)$ (for assets A) are shown in Table 2. As we can verify from Figure 3 that we have $p_1 < 1/2 < p_2$ and $p_1' > 1/2 > p_2'$ for all values of $\mu < 1/2$.

$$
\begin{aligned}
&\text{Given } \mu < 1/2, \text{we have} \\
&p_1 < 1/2 < p_2 \\
&p_1' > 1/2 > p_2'
\end{aligned} \tag{6}
$$

Notice the Table 2 represents the condition when the budget is non-binding i.e. the poor participant is able to spend $\overline{\nu}$. i.e. $\overline{\nu} < y_1 < y_2$. When the poor participant faces a binding constraint i.e. $y_1 < \overline{\nu} < y_2$, then the rich participant knows that the best case probability to win for the poor participant is $\frac{1}{2}$. The rich participant's decision to participate would depend only on utility from savings and the relative advantage from gain in probability. This is elaborated further in Section 3.1.1.

Since the number of poor participants is likely to be higher than that of rich participants in a typical competition, the unequal number of poor and rich participants seems also worth considering in the model. Setting $N = 3, P = 2$ in the equations 1, 2 and 3, for example, we can interpret the **three-player** model with the following values of p_1, p_2, p_1' and p_2'

$$
\begin{aligned}
p_1 &= \frac{\mu}{8(1-\mu)} \\
p_2 &= 1 - 2p_1 \\
p_1' &= \frac{19\mu^2 - 40\mu + 20}{40(1-\mu)^2} \\
p_2' &= 1 - 2p_1'
\end{aligned} \tag{7}
$$

Case	Description	Probability of poor participant to win	Response to μ
I	Only rich invest in skills	$\frac{\mu}{6(1-\mu)}$	With higher μ, the participants with raw-talent get rewarded. The probability to win increases with higher μ for the poorer participant.
II	Only poor invest in skills	$\frac{6-7\mu}{6(1-\mu)}$	With higher μ, the poor participants with access to skills get rewarded. Therefore, The probability to win decreases with higher μ for the poorer participant.
III,IV	Neither or Both invest in skills	$\frac{1}{2}$	Probabilty to win is independent of μ.

Table 3: The probability of winning for the poor participant in the 3-player model

Case	Description	Probability of rich participant to win	Response to μ
I	Only rich invest in skills	$1 - \dfrac{\mu}{6(1-\mu)}$	With higher μ, the participants with raw-talent get rewarded. Therefore, the probability to win decreases with higher μ for the rich participant.
II	Only poor invest in skills	$1 - \dfrac{6-7\mu}{6(1-\mu)}$	With higher μ, the poor participants with access to skills get rewarded. Therefore, the probability to win increases with higher μ for the rich participant.
III,IV	Neither or Both invest in skills	$\dfrac{1}{2}$	Probabilty to win is independent of μ.

Table 4: The probability of winning for the rich participant in the 3-player model

Comparing these probabilities with those from the 2-player model in Equation 5 (also listed in Tables 3 and 4), we see that the probability to win for the poor participant in the 3-player model is lower with respect to the Case III probability (i.e. $1/N$). Likewise, the probability for the rich participant to win is higher in the 3-player model with respect to the Case III probability ($1/N$). The implications for the unequal number of poor and rich participants are further discussed in Section 3.1.2.

3.1.1 Binding constraint

Assuming EUT participants we first consider the simpler case of binding constraint $y_1 < \overline{\nu} < y_2$. Since we assume a stability in long-term participant behaviour (which itself is a result of observed probability of success), mixed-strategies are not relevant in the current set up. Considering pure strategies for participation, the rich participant would thus participate only when the utility from participating would be higher than from not participating i.e. $u(y_2 - \overline{\nu}) + \frac{p_2 u(y_2) + (1-p_2)u(y_1)}{\delta} > u(y_2) + \frac{u(y_2)+u(y_1)}{2\delta}$. This results in the following condition

$$\frac{(p_2 - 1/2)(u(y_2) - u(y_1))}{\delta} > u(y_2) - u(y_2 - \overline{\nu})$$

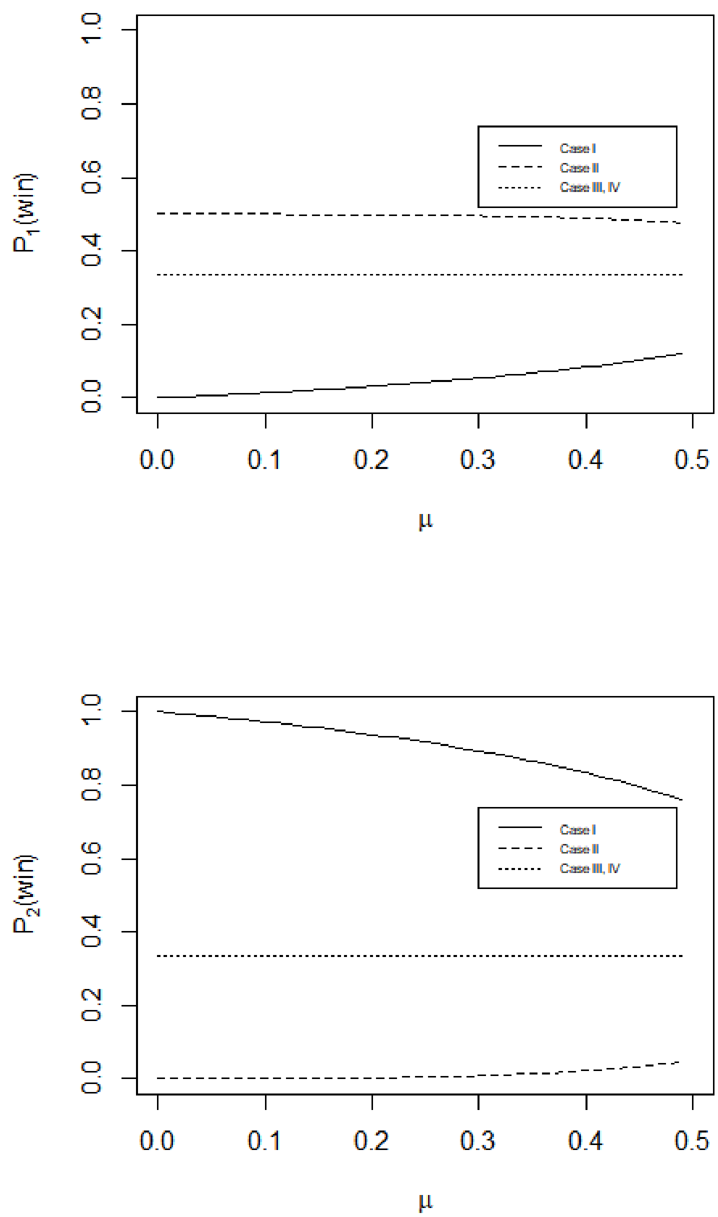

Figure 4: Probability of winning $P_1(\text{win})$, $P_2(\text{win})$ for poor and rich participants with varying μ

With income differences being the same an increase in participation cost $\overline{\nu}$ discourages participation since RHS increases with $\overline{\nu}$ ($\because u(y_2 - \overline{\nu})$ is decreasing in $\overline{\nu}$) and may fail the above condition. Similarly, with participation cost $\overline{\nu}$ being the same, an increase in income-difference $y_2 - y_1$ is in the rich participant's interest whereas a narrowing of income differences reduces incentive for the participant to participate by possibly failing the above condition. All else being the same, higher impatience (lower time-discounted value of money or a higher δ) would also discourage participation. Therefore, when budget is binding for the poor participant i.e. if the poor participant cannot participate due to $y_1 < \overline{\nu}$, then narrower income differences, higher participation costs and higher patience for savings (common for both rich and poor participants) all encourage the rich participant to withdraw from the game. This also means that wider income differences ($y_2 - y_1$) combined with lower participation costs ($\overline{\nu}$) may together help maintain the equilibrium where only the rich participates. Similarly, wider income differences (higher $y_2 - y_1$) and more patience for savings (higher δ) would together maintain the incentive for the rich participant to participate. These conclusions very much fit with the sociological observations made in the literature about status (Hirsch, 1977; Galbraith, 2001).

3.1.2 Non-Binding constraint

When budget is non-binding, then participation of the rich participant is her dominant strategy if the condition is also $u(y_2) + \frac{u(y_2)p_2' + (1-p_2')u(y_1)}{\delta} < u(y_2 - \overline{\nu}) + \frac{u(y_2)+u(y_1)}{2\delta})$ i.e.

$$\frac{(1/2 - p_2')(u(y_2) - u(y_1))}{\delta} > u(y_2) - u(y_2 - \overline{\nu})$$

The inferences on how $y_2 - y_1$, $\overline{\nu}$ and δ influence the participation (see Section 3.1.1) remain the same for the above condition. To discuss the more general Nash equilibria, we now consider the conditions for all of the Cases I, II, III and IV to become a Nash equilibrium. For ease of notation we define the following for a given $\overline{\nu}, y_1$ and y_2

$$U_1 \equiv u(y_1) - u(y_1 - \overline{\nu})$$

$$U_2 \equiv u(y_2) - u(y_2 - \overline{\nu})$$

One could view U_1 and U_2 as the immediate gain in wealth utility from withdrawing. If $u(\cdot)$ represents a diminishing utility, then it is also necessary for $y_1 < y_2$ that

$$U_2 < U_1$$

Consider now that conditions for Case I to be a Nash Equilibrium. If Case I were to be a Nash equilibrium, then for the rich participant's decision to participate, the poor participant must get more utility from withdrawing than from participating and for poor participant's decision to withdraw, the rich must get more utility from participating than from withdrawing. The following conditions [2] must therefore hold simultaneously

[2] $u(y_1) + \frac{p_1 u(y_2) + (1-p_1)u(y_1)}{\delta} > u(y_1 - \overline{\nu}) + \frac{\frac{1}{N}u(y_2) + \frac{1}{N}u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \overline{\nu}) > \frac{(\frac{1}{N} - p_1)(u(y_2) - u(y_1))}{\delta}$ and $u(y_2 - \overline{\nu}) + \frac{p_2 u(y_2) + (1-p_2)u(y_1)}{\delta} > u(y_2) + \frac{\frac{1}{N}u(y_2) + \frac{1}{N}u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \overline{\nu}) < \frac{(p_2 - 1/N)(u(y_2) - u(y_1))}{\delta}$

$$\frac{(\frac{1}{N}-p_1)(u(y_2)-u(y_1))}{\delta} < U_1$$
$$U_2 < \frac{(p_2-1/N)(u(y_2)-u(y_1))}{\delta} \tag{8}$$

Since $p_2 - \frac{1}{N} \geq \frac{1}{N} - p_1$ (strict inequality follows when the number of poor participants P is higher than that for rich participants $N-P$ e.g., with $N=3$ and $P=2$), this is a fairly general condition for the equilibrium. An increase in U_1 due to higher $\overline{\nu}$ does not change the equilibrium conditions so long as the corresponding rise U_2 (due to higher $\overline{\nu}$) does not affect the second inequality. In other words, a higher $\overline{\nu}$ favours the equilibrium (Case I) so long as there is an incentive for the rich to participate. Likewise, a decrease in $\overline{\nu}$ would be favourable for the equilibrium (Case I) so long as the poor participants do not also have an incentive to participate.

Similar effects can be seen with income differences $y_2 - y_1$ (which increase $u(y_2)-u(y_1)$ as well as $\frac{U_2}{U_1}$) if we rewrite Equation 8 as

$$\frac{(\frac{1}{N}-p_1)}{\delta} < \frac{U_1}{u(y_2)-u(y_1)}$$
$$\frac{U_2}{(u(y_2)-u(y_1))} < \frac{(p_2-1/N)}{\delta}$$

Limiting our attention to cases when $\overline{\nu} < y_1$, higher y_2 for fixed y_1 decreases $\frac{U_2}{u(y_2)-u(y_1)}$ and makes it more likely for the rich to participate while it also decreases $\frac{U_1}{u(y_2)-u(y_1)}$ by raising the stakes enough for the poor to participate (making it less likely for first inequality to be fulfilled). It is easy to see that higher impatience δ encourages both poor and rich to withdraw. As we shall see, equilibrium conditions corresponding to Case II are far more stringent.

If Case II is a Nash Equilibrium, then for rich's decision to withdraw, the poor participant should get more utility from participating than from withdrawing and for poor participant's decision to participate, the rich participant should get more utility from withdrawing than from participating. This means that the following hold [3] simultaneously,

$$U_1 < \frac{(p_1'-1/N)(u(y_2)-u(y_1))}{\delta}$$
$$U_2 > \frac{(1/N-p_2')(u(y_2)-u(y_1))}{\delta}$$

Since $\frac{(p_1'-1/N)(u(y_2)-u(y_1))}{\delta} < \frac{(1/N-p_2')(u(y_2)-u(y_1))}{\delta}$, the above cannot be satisfied so long as we have a diminishing utility which necessitates $U_2 < U_1$. Thus an equilibrium with Case II which incentivises the poor to participate but not the rich is possible only with a reference dependent utility with $U_1 < U_2$. With $U_1 < U_2$, we can write above as

$$\frac{U_1}{u(y_2)-u(y_1)} < \frac{(p_1'-1/N)}{\delta} < \frac{(1/N-p_2')}{\delta} < \frac{U_2}{u(y_2)-u(y_1)}$$

[3] $u(y_1-\overline{\nu}) + \frac{p_1' u(y_2)+(1-p_1')u(y_1)}{\delta} > u(y_1) + \frac{\frac{1}{N}u(y_2)+\frac{1}{N}u(y_1)}{\delta} \Rightarrow u(y_1)-u(y_1-\overline{\nu}) < \frac{(p_1'-1/N)(u(y_2)-u(y_1))}{\delta}$ and $u(y_2) + \frac{p_2' u(y_2)+(1-p_2')u(y_1)}{\delta} > u(y_2-\overline{\nu}) + \frac{\frac{1}{N}u(y_2)+\frac{1}{N}u(y_1)}{\delta} \Rightarrow u(y_2)-u(y_2-\overline{\nu}) > \frac{(1/N-p_2')(u(y_2)-u(y_1))}{\delta}$

Notice that despite a reference-dependent utility, a higher $\overline{\nu}$ or more impatience (δ) may disincentivise the poor from participation just the same (leading to Case III). Similarly, higher income differences (an increase in y_2 for fixed y_1) could encourage the rich to participate by decreasing $\frac{U_2}{u(y_2)-u(y_1)}$ (leading to Case IV). Even with the reference dependent utility, the equilibrium with Case II remains a fragile one.

If Case III were to be a Nash-Equilibrium, then for rich participant's decision to withdraw, poor should get more utility from withdrawing than from participating and for poor participant's decision to withdraw, the rich participant should get more from withdrawing than from participating. The following should thus hold[4],

$$U_1 > \frac{(p_1' - \frac{1}{N})(u(y_2) - u(y_1))}{\delta}$$
$$U_2 > \frac{(p_2 - \frac{1}{N})(u(y_2) - u(y_1))}{\delta}$$

Since $p_1' - \frac{1}{N} \leq p_2 - \frac{1}{N}$ (strict inequality when number of poor participants is higher), we have $\frac{(p_1'-\frac{1}{N})(u(y_2)-u(y_1))}{\delta} < \frac{(p_2-\frac{1}{N})(u(y_2)-u(y_1))}{\delta}$ and $U_2 < U_1$. A higher $\overline{\nu}$ pushes both U_2 and U_1 further away from participation but a fall in $\overline{\nu}$, incentivises the rich participant to participate (leading to Case I).

$$\frac{U_1}{u(y_2) - u(y_1)} > \frac{(p_1' - \frac{1}{N})}{\delta}$$
$$\frac{U_2}{u(y_2) - u(y_1)} > \frac{(p_2 - \frac{1}{N})}{\delta}$$

Higher income differences i.e. higher y_2 for fixed y_1 also decrease $\frac{U_2}{u(y_2)-u(y_1)}$ and incentivise the rich participant to participate (leading to an equilibrium with Case I). Conversely, if the game is not lucrative for the rich it would automatically be infeasible for the poor i.e. if rich withdraws then poor automatically withdraws. If there is any net utility from playing the game, however, Case IV is the likely equilibrium.

If Case IV were to be a Nash Equilibrium, then for rich's decision to participate, the poor participant should get more from participating than from withdrawing and for poor's decision to participate, the rich participant should get more from participating than from withdrawing. Therefore, the following[5] must hold

$$U_1 < \frac{(\frac{1}{N} - p_1)(u(y_2) - u(y_1))}{\delta}$$
$$U_2 < \frac{(\frac{1}{N} - p_2')(u(y_2) - u(y_1))}{\delta}$$

With $\frac{(\frac{1}{N}-p_2')(u(y_2)-u(y_1))}{\delta} < \frac{(\frac{1}{N}-p_1)(u(y_2)-u(y_1))}{\delta}$ and $U_2 < U_1$, this equilibrium rests upon the incentive for the poor participant to participate. So long as the poorer participant finds the competition worthwhile, the rich would also participate.

---

[4] $u(y_1) + \frac{\frac{1}{N}u(y_2)+\frac{1}{N}u(y_1)}{\delta} > u(y_1 - \overline{\nu}) + \frac{p_1'u(y_2)+(1-p_1')u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \overline{\nu}) > \frac{(p_1'-\frac{1}{N})(u(y_2)-u(y_1))}{\delta}$ and $u(y_2) + \frac{\frac{1}{N}u(y_2)+\frac{1}{N}u(y_1)}{\delta} > u(y_2 - \overline{\nu}) + \frac{p_2u(y_2)+(1-p_2)u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \overline{\nu}) > \frac{(p_2-\frac{1}{N})(u(y_2)-u(y_1))}{\delta}$

[5] $u(y_1 - \overline{\nu}) + \frac{\frac{1}{N}u(y_2)+\frac{1}{N}u(y_1)}{\delta} > u(y_1) + \frac{p_1u(y_2)+(1-p_1)u(y_1)}{\delta} \Rightarrow u(y_1) - u(y_1 - \overline{\nu}) < \frac{(\frac{1}{N}-p_1)(u(y_2)-u(y_1))}{\delta}$ and $u(y_2 - \overline{\nu}) + \frac{\frac{1}{N}u(y_2)+\frac{1}{N}u(y_1)}{\delta} > u(y_2) + \frac{p_2'u(y_2)+(1-p_2')u(y_1)}{\delta} \Rightarrow u(y_2) - u(y_2 - \overline{\nu}) < \frac{(\frac{1}{N}-p_2')(u(y_2)-u(y_1))}{\delta}$

The above results outline how participation costs influence the investments towards status. For very low $\overline{\nu}$, rich and poor would both participate and for very high $\overline{\nu}$, both would withdraw. For everything in between, the only equilibrium with a diminishing utility from wealth is one where rich participate whereas poor don't.

The above framework also allows us to discuss the implications of inequality for the same total wealth in the economy. As such, a segregation where the poor never participate in the status economy seems less favourable than one where both rich and poor participate in the status competitions. The current chapter explains that the likely outcome of a failure to maintain universal participation in status games (under diminishing utility from wealth) is that of a segregation where only the rich participate in skills leading to economic success. While a reference-based utility (which may be encouraged with segregation) does seem to permit an equilibrium where only the poor are concerned with status games (with respect to skills-enhancement), the result only suggests that a segregation of status games along with that of incomes is another possible long-term outcome of limited participation.

4 Conclusion

We have considered a model where status amounts to a position that only temporarily cements the participant's economic position. With a decision theory framework for uncertainty, the goal of the model has been to investigate role of income difference on expenditures that carry some status while also having some effect on the income mobility of the participants. The endogenisation of status in this manner allows an understanding of the role of income difference in status consumption in a generic framework.

The foremost result from the model is that under diminishing utility from wealth, the only two stable equilibria are those where both rich and poor participate in status competitions and those where only rich stand to benefit from status competitions. This explains the limits to the poor offered by participation in status competitions.

The model also allows us to understand how a withdrawal from expenditure such as education (particularly in certain developing economies where income differences could be larger) could be the result of a combined effect of both participation costs and income-differences. More specifically, high participation costs push the equilibrium to one where where only rich participate in status-related competitions while high income differences may still have the unlikely effect of incentivising the poor participants to participate. Lowering of participation costs in education (as a status game) despite rise in income differences may thus also represent a likely escape from segregation of participation in the near-term.

Lastly, the role of a reference-based utility adds an important caveat to these results since contentment at lower levels of wealth could support segregation of competitions at different income levels in the economy. This is also in line with many of the findings from subjective contentment data on the belief in meritocracy and persisting inequalities in the developed economies.

References

Adler, I., & Kraus, V. (1985). Components of occupational prestige evaluations. *Work and Occupations*, *12*(1), 23–39.

Chai, A., Kaus, W., & Kiedaisch, C. (2019, July). Conspicuous spending and the income distribution of social groups. *Economic Inquiry*, *57*(3), 1324-1341.

Charles, K. K., Hurst, E., & Roussanov, N. (2009). Conspicuous consumption and race. *The Quarterly Journal of Economics*. (http://www.nber.org/papers/w13392)

Cole, H. L., Mailath, G. J., & Postlewaite, A. (1992). Social norms, savings behavior, and growth. *Journal of Political Economy*, *100*(6), 1092–1125. Retrieved from `http://www.jstor.org/stable/2138828`

Coleman, J. (1990). *Foundations of social theory*. Harvard University Press.

Falk, A., & Knell, M. (2004). Choosing the joneses: Endogenous goals and reference standards. *The Scandinavian Journal of Economics*, *106*(3), 417–435.

Frank, R. H. (1993). *Choosing the right pond*. OUP USA.

Friehe, T., & Mechtel, M. (2014). Conspicuous consumption and political regimes: Evidence from east and west Germany. *European Economic Review*.

Galbraith, J. K. (1958). *The affluent society*.

Galbraith, J. K. (1994). Beyond contentment. *The Telegraph*, *14*.

Galbraith, J. K. (2001). *The Essential Galbraith* (A. D. Williams, Ed.). Houghton Mifflin Company.

Goldthorpe, J. H. (2000). *On sociology: Numbers, narratives, and the integration of research and theory*. Oxford University Press, USA.

Goldthorpe, J. H., & Hope, K. (1974). *The social grading of occupations. a new scale and approach*. Clarendon Press.

Goode, W. J. (1978). *The celebration of heroes: Prestige as a social control system*. Univ of California Press.

Haller, M., Jowell, R., & Smith, T. W. (2009). *The international social survey programme 1984-2009: Charting the globe*. Routledge.

Heffetz, O. (2011). A test of conspicuous consumption: Visibility and income elasticities. *The Review of Economics and Statistics*, *XCIII*(4).

Hirsch, F. (1977). *Social limits to growth*. Routledge and Kegan Paul Ltd.

Ireland, N. J. (1994). On limiting the market for status signals. *Journal of Public Economics*.

Jaikumar, S., & Sarin, A. (2015). Conspicuous consumption and income inequality in an emerging economy: evidence from India. *Marketing Letters*.

Jin, Y., Li, H., & Wu, B. (2011). Income inequality, consumption, and social-status seeking. *Journal of Comparative Economics*, *39*(2), 191–204.

Kaus, W. (2013). Conspicuous consumption and "race": Evidence from South Africa. *Journal of Development Economics*, *100*(1), 63-73. Retrieved from `https://www.sciencedirect.com/science/article/pii/S0304387812000648`

Khamis, M., Prakash, N., & Siddique, Z. (2012). Consumption and social identity: Evidence from India. *Journal of Economic Behavior and Organization*, *83*(3), 353-371. Retrieved from `https://www.sciencedirect.com/science/article/pii/S0167268112001345`

Knell, M. (1999). Social comparisons, inequality, and growth. *Journal of Institutional and Theoretical Economics (JITE) / Zeitschrift fÃŒr die gesamte Staatswissenschaft*, *155*(4), 664–695.

Mazzocco, P. J., Rucker, D. D., Galinsky, A. D., & Anderson, E. T. (2012). Direct and vicarious conspicuous consumption: Identification with low-status groups increases the desire for high-status goods. *Journal of Consumer Psychology*, *22*(4), 520–528. Retrieved from `http://www.jstor.org/stable/26578155`

Mijs, J. J., & Savage, M. (2020). Meritocracy, elitism and inequality. *The Political Quarterly*, *91*(2), 397-404.
Mijs, J. J. B. (2019, 01). The paradox of inequality: income inequality and belief in meritocracy go hand in hand. *Socio-Economic Review*, *19*(1), 7-35.
Moldovanu, B., Sela, A., & Shi, X. (2007). Contests for status. *Journal of political Economy*, *115*(2), 338–363.
Nieftagodien, S., Van Der Berg, S., et al. (2007). Consumption patterns and the black middle class: The role of assets. *Bureau of Economic Research*, *2*, 1–10.
Omori, M., & Smith, D. T. (2015). The impact of socioeconomic status and metropolitan area racial composition on visible consumption among whites and racial/ethnic minorities. *Race Soc Probl*(7), 169-180.
Postlewaite, A. (1998). The social basis of interdependent preferences. *European economic review*, *42*(3-5), 779–800.
Roychowdhury, P. (2016). Visible inequality, status competition, and conspicuous consumption: evidence from rural India. *Oxford Economic Papers*, *69*(1), 36–54.
Truyts, T. (2010). Social status in economic theory. *Journal of Economic Surveys*, *24*(1), 137–169.
Veblen, T. (1899). *Theory of the leisure class*.
Wegener, B. (1992). Concepts and measurement of prestige. *Annual review of sociology*, *18*(1), 253–280.
Wrong, D. (1979). *Power: Its forms, bases and uses*. Routledge.

Appendices

A Convolution of Uniform Random Variables

The distribution of a convex combination $w = \mu r + (1 - \mu)s$ of two random variables r, s - which are distributed according to densities $f_R(r)$ and $f_S(s)$ - can be derived using the convolution of two derived variables $x = \mu r$ and $y = (1 - \mu)s$. If r and s are uniformly distributed variables ranging from 0 to 1, then x ranges from 0 to μ and y ranges from 0 to $(1 - \mu)$. The CDF for $F_X(x)$ and $F_Y(y)$ are simply $F_X(x) = \frac{x}{\mu}$, $F_Y(y) = \frac{y}{1-\mu}$ (respectively). Therefore, the densities for x and y are $f_X(x) = \frac{1}{\mu}$ and $f_Y(y) = \frac{1}{1-\mu}$. The density of w is simple $f(w) = \int_{-\infty}^{\infty} f_Y(w - x)f_X(x)dx$. Since , $f_X(x)$ is $1/\mu$ only between 0 and μ. We have the following integral to evaluate for the convolution

$$f(w) = \frac{1}{\mu}\int_0^{\mu} f_Y(w - x)dx$$

Since $f_Y(y)$ is 1 only between 0 and $1 - \mu$, we have the condition $0 < w - x < 1 - \mu \Rightarrow w - (1 - \mu) < x < w$. Further, x is distributed only between 0 and μ. As shown in Figure 5, the integral would evaluate to the following[6] under the assumption $\mu < \frac{1}{2}$.

[6]Notice that as a convex combination, w is distributed only between 0 and 1.

$$f(w) = \frac{1}{\mu}\int_0^{\mu} f_Y(w-x)dx = \begin{cases} \frac{1}{\mu}\int_0^{w} \frac{1}{1-\mu}dx & 0 < w < \mu \\ \frac{1}{\mu}\int_0^{\mu} \frac{1}{1-\mu}dx & \mu < w < 1-\mu \\ \frac{1}{\mu}\int_{w-(1-\mu)}^{\mu} \frac{1}{1-\mu}dx & 1-\mu < w < 1 \\ 0 & \text{otherwise} \end{cases}$$

This can be simplified as

$$f(w) = \begin{cases} \frac{w}{\mu(1-\mu)} & 0 < w < \mu \\ \frac{1}{1-\mu} & \mu < w < 1-\mu \\ \frac{1-w}{\mu(1-\mu)} & 1-\mu < w < 1 \\ 0 & \text{otherwise} \end{cases}$$

Figure 6 shows this trapezoidal density with μ set to $\frac{1}{5}$. The CDF $F_X(x)$ can be evaluated by integrating the above density as $P(X < x) = \int_{-\infty}^{x} f(t)dt$.

$$F_X(x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{\mu}{2(1-\mu)} + \frac{x-\mu}{1-\mu} & \mu < x < 1-\mu \\ \frac{\mu^2}{2\mu(1-\mu)} + \frac{2\mu(1-2\mu)}{2\mu(1-\mu)} + \frac{\mu^2-x^2+2x-1}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Simplifying above, we have

$$F_X(x) = \begin{cases} \frac{x^2}{2\mu(1-\mu)} & 0 < x < \mu \\ \frac{x-\mu/2}{1-\mu} & \mu < x < 1-\mu \\ 1 - \frac{(x-1)^2}{2\mu(1-\mu)} & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

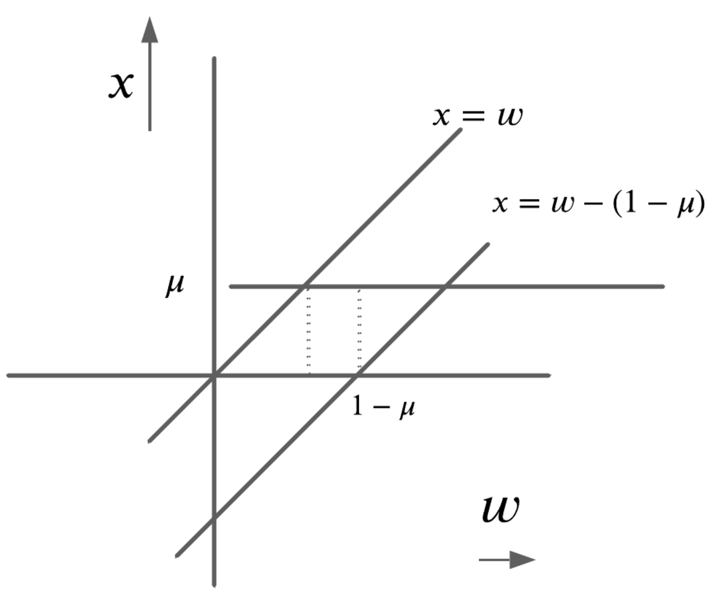

Figure 5: Ranges of w and $w - x$ for convolution of two scaled random variables

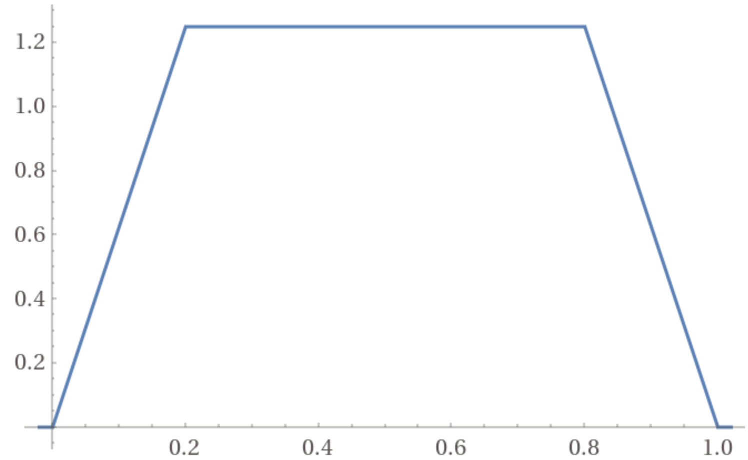

Figure 6: Combined density of w as the convex combination of two random variables with $\mu = \frac{1}{5}$

B $P(\textbf{win}|r, s)$

To calculate $\int_{s=0}^{s=1}\int_{r=0}^{r=1} P(\text{win}|r,s)drds$ over the variable x (where $0 < r < 1$ and $0 < s < 1$), we evaluate the sum of three parts split by the boundaries of the piecewise function. The three parts are shown in Figure 7. If $x = \mu r + (1-\mu)s$ and $F(x)$ is of the following piece-wise form

$$F(x) = \begin{cases} A & 0 < x < \mu \\ B & \mu < x < 1-\mu \\ C & 1-\mu < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Then the integral $\int_{s=0}^{s=1}\int_{r=0}^{r=1} F(x)drds$ can be written as

$$\int_{s=0}^{s=1}\int_{r=0}^{r=1} F(x)dsdr = \int_{r=0}^{r=1}\int_{s=0}^{s=\frac{\mu(1-r)}{1-\mu}} A\, dsdr + \int_{r=0}^{r=1}\int_{s=\frac{\mu(1-r)}{1-\mu}}^{s=1-\frac{\mu r}{1-\mu}} B\, dsdr + \int_{r=0}^{r=1}\int_{s=1-\frac{\mu r}{1-\mu}}^{s=1} C\, dsdr$$

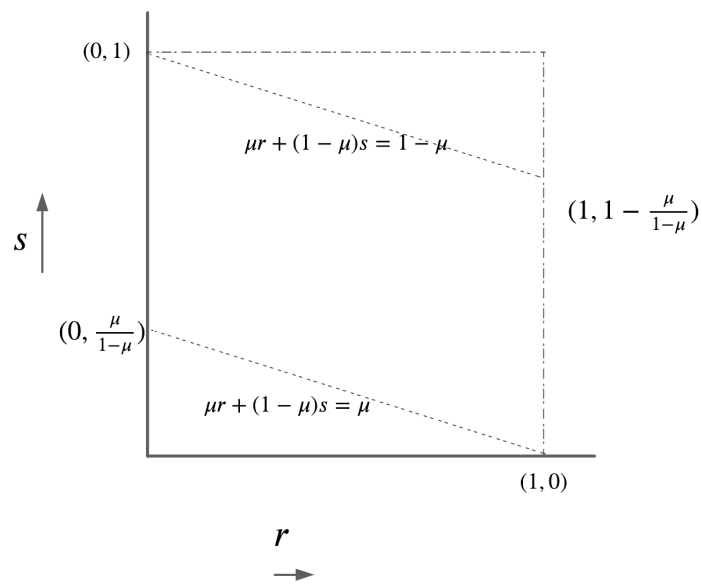

Figure 7: Integration over r, s split by boundaries $\mu r + (1-\mu)s = \mu$ and $\mu r + (1-\mu)s = 1-\mu$